

Student Takeaway for Sync Session

<<Week 17: Support Vector Machines for Classification - Part I>>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Support Vector Machines for Classification - Part I		
Session duration	2 hours		
Session type	Lectures: Conceptual understanding of discriminant functions, Gaussian methods, Perceptron, Logistic Regression, and GMM. Case Studies / Demos: Application of these concepts to real-world datasets like customer churn and predictive maintenance.		
Scope	This session explores discriminant functions for classification. It covers: - How algorithms form linear and nonlinear decision boundaries. - The theory behind Gaussian contours, Perceptron, Logistic Regression, and GMM. - Practical demos visualising these concepts on real datasets.		
Learning objectives	Objective		Core capability
	Understand Decision Boundaries		Relate discriminant function theory to real-world data visuals
	Explore Gaussian Contours		Interpret class overlap and uncertainty using probability densities
	Compare Linear & Nonlinear Discriminants		Evaluate when to use simple vs. complex boundary models
	Apply GMM for Complex Data		Use mixture models for advanced clustering and classification
Software/Tools	- Python (Libraries for Gaussian methods, Perceptron, Logistic Regression, GMM) - IDE: Jupyter Notebook - Datasets: Churn Data, Predictive Maintenance Data		

Pro tips for success:

Ask Bold Questions: Clarify how different models create boundaries.

Be Hands-On: Try running the demo code yourself.

Collaborate: Share your insights on model selection for different scenarios.

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-3	Session Overview	Introduction to the session's focus on understanding decision boundaries through demos.	
4	Concept Recap	Review core principles: Discriminant Functions, Gaussian Methods, Perceptron, Logistic Regression, GMM.	Key theory from async materials
5	Demo 1: Gaussian Contours	Visualise class overlap on churn data using Gaussian statistics and probability densities.	Customer Churn Data
6	Demo 2: Predictive Maintenance	Compare Linear (Logistic Regression) vs. Nonlinear (KNN) discriminants.	Predictive Maintenance Data
7	Demo 3: Perceptron	Demonstrate the Perceptron as a fast, linear classifier on churn data.	Linearly separable data
8	Demo 4: GMM	Introduce GMM for creating nonlinear boundaries on complex, multi-modal data.	Churn segmentation
9	Recap and Reflection	Comparative analysis of the discriminant models covered.	
10	Discussion	Discussion on choosing and evaluating discriminant models.	
11	Q&A Session	Recap and open floor for questions and clarifications.	

Post session

Reflection challenge	How does the choice of a linear vs. nonlinear model impact the interpretation of a decision boundary in your domain?
Explore more	Read: Research papers on applications of GMM. Experiment: Apply Logistic Regression and a nonlinear model to a simple dataset and compare the decision boundaries.
Get inspired	Many classification systems in finance, healthcare, and marketing rely on the techniques you've seen.
The journey ahead	Prepare for our next session, where we'll build on these foundational concepts.